

Packet Size and Overhead

Some readers will worry about the increase in packet size due to new headers, padding, and the Message Authentication Code (MAC). The minimum packet size is in the order of 28 bytes (depending on negotiated algorithms). The increase is negligible for large packets, but very significant for one-byte packets (telnet-type sessions). There are, however, several factors that make this a non-issue in almost all cases:

- The minimum size of a TCP/IP header is 32 bytes. Thus, the increase is actually from 33 to 51 bytes (roughly).

- The minimum size of the data field of an Ethernet packet is 46 bytes [RFC0894]. Thus, the increase is no more than 5 bytes. When Ethernet headers are considered, the increase is less than 10 percent.

- The total fraction of telnet-type data in the Internet is negligible, even with increased packet sizes.

The only environment where the packet size increase is likely to have a significant effect is PPP [RFC1661] over slow modem lines (PPP compresses the TCP/IP headers, emphasizing the increase in packet size). However, with modern modems, the time needed to transfer is in the order of 2 milliseconds, which is a lot faster than people can type.

There are also issues related to the maximum packet size. To minimize delays in screen updates, one does not want excessively large packets for interactive sessions. The maximum packet size is negotiated separately for each channel.

Binary Packet Protocol

Each packet is in the following format:

uint32 packet_length

byte padding_length

byte[n1] payload; n1 = packet_length - padding_length - 1

byte[n2] random padding; n2 = padding_length

byte[m] mac (Message Authentication Code - MAC); m = mac_length

packet_length

The length of the packet in bytes, not including 'mac' or the 'packet_length' field itself.

padding_length

Length of 'random padding' (bytes).

payload

The useful contents of the packet. If compression has been negotiated, this field is compressed. Initially, compression MUST be "none".

random padding

Arbitrary-length padding, such that the total length of (packet_length || padding_length || payload || random padding) is a multiple of the cipher block size or 8, whichever is larger. There MUST be at least four bytes of padding. The padding SHOULD consist of random bytes. The maximum amount of padding is 255 bytes.

mac

Message Authentication Code. If message authentication has been negotiated, this field contains the MAC bytes. Initially, the MAC algorithm MUST be "none".

Symbolic name	reason code
SSH_DISCONNECT_HOST_NOT_ALLOWED_TO_CONNECT	1
SSH_DISCONNECT_PROTOCOL_ERROR	2
SSH_DISCONNECT_KEY_EXCHANGE_FAILED	3
SSH_DISCONNECT_RESERVED	4
SSH_DISCONNECT_MAC_ERROR	5
SSH_DISCONNECT_COMPRESSION_ERROR	6
SSH_DISCONNECT_SERVICE_NOT_AVAILABLE	7
SSH_DISCONNECT_PROTOCOL_VERSION_NOT_SUPPORTED	8
SSH_DISCONNECT_HOST_KEY_NOT_VERIFIABLE	9
SSH_DISCONNECT_CONNECTION_LOST	10
SSH_DISCONNECT_BY_APPLICATION	11
SSH_DISCONNECT_TOO_MANY_CONNECTIONS	12
SSH_DISCONNECT_AUTH_CANCELLED_BY_USER	13
SSH_DISCONNECT_NO_MORE_AUTH_METHODS_AVAILABLE	14
SSH_DISCONNECT_ILLEGAL_USER_NAME	15

<https://datatracker.ietf.org/doc/html/rfc2616>